1. $z = 7 - 9i = \sqrt{130}\, e^{i(2\pi - \operatorname{arctg}\frac{9}{7})}$

$|z| = \sqrt{49 + 81} = \sqrt{130}$

$\operatorname{Arg} z = \operatorname{arcctg} -\frac{9}{7} = 2\pi - \operatorname{arcctg}\frac{9}{7}$

2. $9e^{i\frac{\pi}{4}} = 9\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$

3. $\frac{(-6+3i)}{-2-i} - \frac{-2-i}{(-6+3i)^2} = \left(\frac{9}{5} + \frac{18}{2025}\right) + i\left(\frac{99}{2025} - \frac{12}{5}\right)$

(1) $\frac{-6+3i}{-2-i} = \frac{(-6+3i)(-2+i)}{(-2-i)(-2+i)} = \frac{12-3+i(-6-6)}{5} = \frac{9-12i}{5} = \frac{9}{5} - \frac{12}{5}i$

(2) $\frac{2+i}{36-9+36i} = \frac{2+i}{27+36i} = \frac{(2+i)(27-36i)}{729+1296} = \frac{54-36i+i(27+72)}{2025} =$

$= \frac{18+99i}{2025} = \frac{18}{2025} + \frac{99i}{2025}$

4. $z = -3 - 2i$; $\bar{z} = -3 + 2i$

$z = -1 - 7i$

$$z^{-1} = \frac{1}{-1-7i} = \frac{-1+7i}{50} = \underline{\underline{-\frac{1}{50} + \frac{7}{50}}}$$

$$\sqrt[5]{\frac{4-i}{2-3i}} = -\sqrt[5]{\frac{(4+i)(2+3i)}{13}} = -\sqrt[5]{\frac{5}{13} + \frac{14}{13}i} = z$$

$$p$$

$$|p| = \sqrt{\frac{25}{169} + \frac{196}{169}} = \frac{\sqrt{221}}{13}$$

$$\arg p = \operatorname{tg}^{-1}\frac{14}{5} = \varphi$$

$$\sqrt[5]{p} = \sqrt[5]{\frac{\sqrt{221}}{13}}\cos\left(\frac{\varphi}{5} + \frac{2\pi}{5}k\right) + i\sqrt[5]{\frac{\sqrt{221}}{13}}\sin\left(\frac{\varphi}{5} + \frac{2\pi}{5}k\right), \quad k \in [0;5) \quad k \in \mathbb{Z}$$

$$z = -\sqrt[5]{p}$$

Ex. $(19+5i)(-6+5i) = (-114-25)+i(-30+95) = -139+65i$

$$\ell\,(15(\cos\tfrac{28}{16}\pi + i\sin\tfrac{28}{16}\pi))\cdot 2(\cos\tfrac{18}{16}\pi + i\sin\tfrac{18}{16}\pi) =$$
$$= 30\,((\cos\tfrac{28}{16}\pi\cos\tfrac{18}{16}\pi - \sin\tfrac{28}{16}\pi\sin\tfrac{18}{16}\pi) + i(\sin\tfrac{28}{16}\pi\cos\tfrac{18}{16}\pi +$$
$$+ \cos\tfrac{28}{16}\pi\sin\tfrac{18}{16}\pi)) = 30(\cos\tfrac{46}{16}\pi + i\sin\tfrac{46}{16}\pi) =$$
$$= 30(\cos\tfrac{23}{8}\pi + i\sin\tfrac{23}{8}\pi) = 30(\cos\tfrac{7}{8}\pi + i\sin\tfrac{7}{8}\pi)$$

8. $z = -5 + 3i$

$|z| = \sqrt{34} \approx 5,83$

$\arg z = \pi - \operatorname{tg}^{-1} \frac{3}{5} \approx 2,6$

$$\begin{cases}(4-3i)z+(-3+i)w=-3+8i\\(-3-i)z+(4+3i)w=1+2i\end{cases}$$

$$\frac{(4-3i)(4+3i)}{-3+i}z+(4+3i)w=\frac{(-3+3i)(4+3i)}{-3+i}$$

$$\frac{25}{10}(-3-i)z+(4+3i)w=\frac{3(-1+i)(4+3i)}{-3+i}$$

$$\frac{5}{2}(-3-i)z+(4+3i)w=\frac{3(-4-3+i)}{-3+i}=$$

$$=\frac{3(-7+i)(-3-i)}{10}=\frac{3}{10}(21+1-8i+7i)=\frac{3}{10}\cdot 5(11+2i)$$

$$\begin{cases}\frac{5}{2}(-3-i)z+(4+3i)w=\frac{3}{5}(11+2i)\\-(-3-i)z+(4+3i)w=1+2i\end{cases}$$

$$\frac{3}{2}(-3-i)z=\underbrace{\frac{33}{5}-\frac{5}{5}}_{\frac{28}{5}}+\underbrace{\frac{6}{5}i-\frac{10}{5}i}_{-\frac{4}{5}}=\frac{28}{5}-\frac{4}{5}i$$

$$15(-3-i)z=56-8i$$

$$z=\frac{(56-8i)(-3+i)}{150}=\frac{8}{150}\cdot 75(7-i)(-3+i)$$

$$z=\frac{4}{75}(-21+1+3i+7i)=\frac{4}{75}\cdot 5(-4+2i)=\frac{4(-20+10i)}{15}=-\frac{16}{15}+\frac{8}{15}i$$

$$w=\frac{(1+2i)+(3+i)z}{(4+3i)}=\frac{((1+2i)+(3+i)z)(4-3i)}{25}$$

- $(3+i)\left(-\frac{16}{15}+\frac{8}{15}i\right)=-\frac{48}{15}-\frac{8}{15}-\frac{16}{15}i+\frac{24}{15}i=-\frac{56}{15}+\frac{8}{15}i$
- $1+2i-\frac{56}{15}+\frac{8}{15}i=-\frac{41}{15}+\frac{38}{15}i$
- $\frac{1}{15}(-41+38i)(4-3i)=\frac{1}{15}\underbrace{(-164+114)}_{-50}+i\underbrace{(152+123)}_{275}=$

$$=-\frac{50}{15\cdot 3}^{10}+\frac{55}{3}i$$

$$w=-\frac{40}{25\cdot 3}^{2}+\frac{55}{25\cdot 3}^{11}i=-\frac{2}{15}+\frac{11}{15}i$$